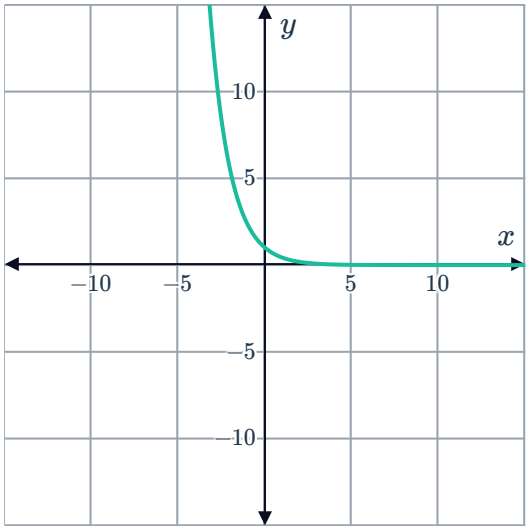
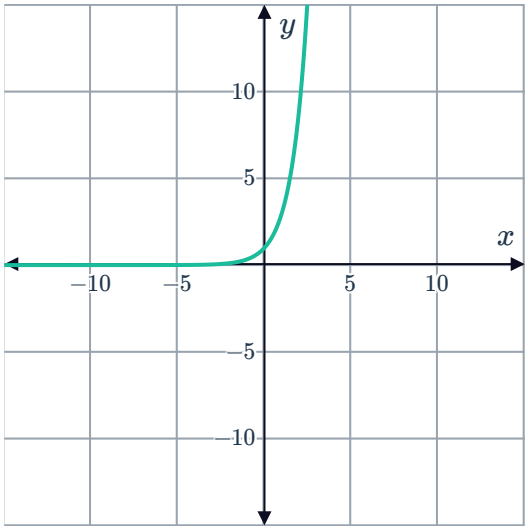
Exponential graphs

1 State whether the following graphs show an exponential function:

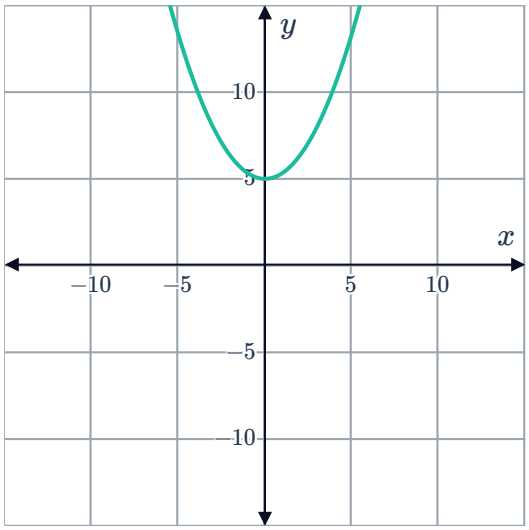
a



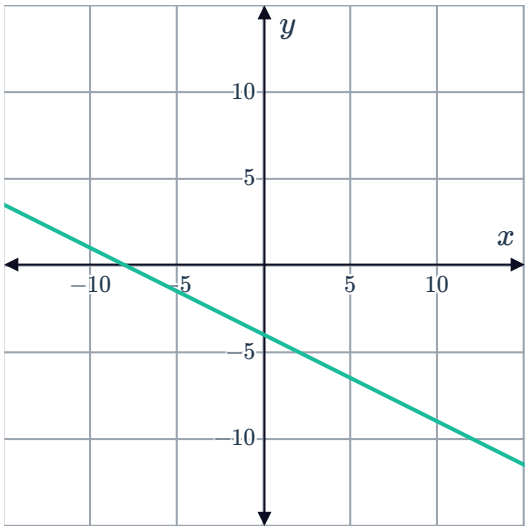
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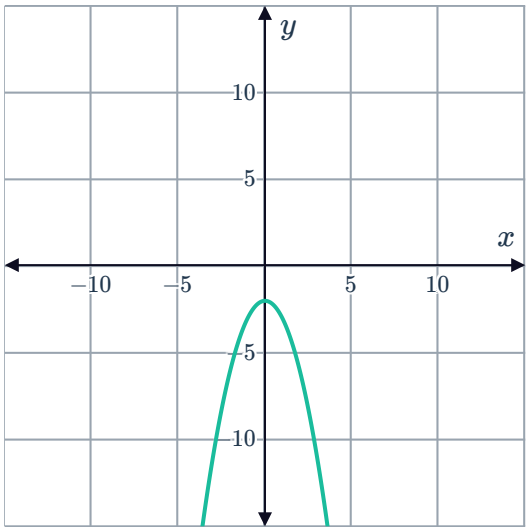
c



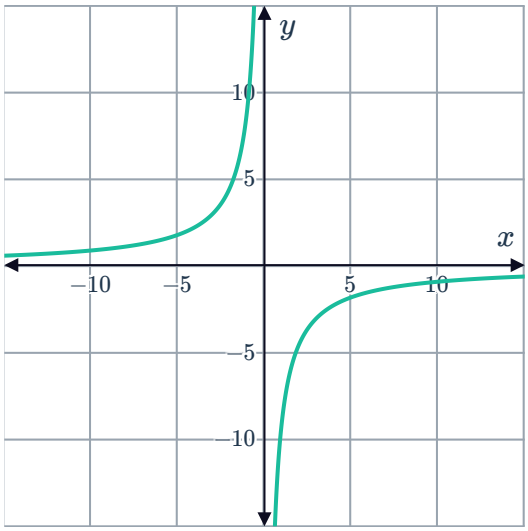
d



e



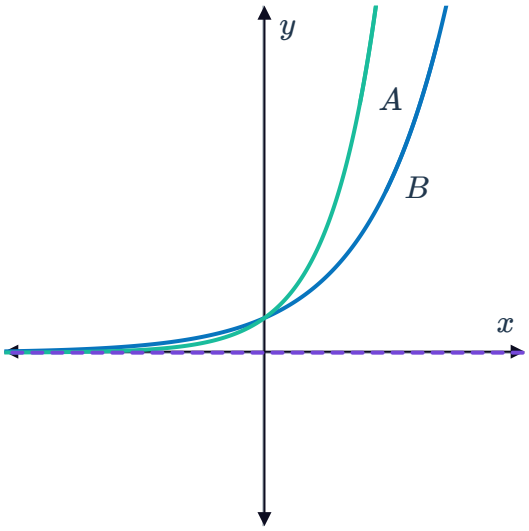
f



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2 Consider the graphs of the two exponential functions A and B:

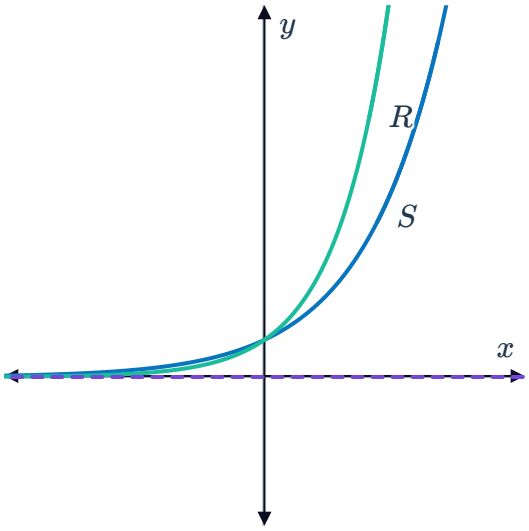
One of the graphs is of $y = 3^x$ and the other graph is of $y = 6^x$. Which is the graph of $y = 3^x$?



3 Consider the graphs of the two exponential functions R and S:

a One of the graphs is of $y = 3^x$ and the other graph is of $y = 5^x$. Which is the graph of $y = 5^x$?

b Which function has its graph below that of the other when $x < 0$? Explain your answer.



4 Determine whether following are increasing or decreasing exponential functions:

a $y = 2^x$

b $y = 2^{-x}$

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Increasing exponential graphs

5 Consider the expression $y = 4^x$.

i Complete the table of values:

x	-3	-2	-1	0	1	2	3
y							

ii Sketch the graph.

iii Describe what happens to the y-values as x increases.

iv Describe what happens to the y-values as x decreases.

v Does the curve cross the x-axis?

vi At what value of y does the graph cross the y-axis?

g Use your graph to approximate the solution of the equation $4^x = 3$.

6 For each of the following functions:

i Find the y-value of the y-intercept of the curve.

ii Complete the table of values:

x	-3	-2	-1	0	1	2	3
y							

iii Find the horizontal asymptote of the curve.

iv Sketch the graph.

a $y = 5^x$ **b** $y = 10^x$ **c** $y = 2^x$

7 Consider the expression $y = 10^x$.

i Can the value of y ever be negative?

ii Can the value of y ever be equal to 0?

iii As the values of x get larger and larger, what value does y approach?

iv As the values of x get smaller and smaller, what value does y approach?

v Find the y-value of the y-intercept of the curve.

vi How many x-intercepts does the curve have?

vii Sketch the graph.

h Use your graph to approximate the solution of the equation $10^x = 2$.

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8 Consider the exponentials $y = 2^x$ and $y = 3^x$:

a Find the horizontal asymptote of each curve:

i $y = 2^x$ **ii** $y = 3^x$

b Complete a table of values for each exponetial:

x	-3	-2	-1	0	1	2	3
y							

i $y = 2^x$ **ii** $y = 3^x$

c Sketch the graphs of $y = 2^x$ and $y = 3^x$ on the same set of axes.

d Find the coordinates of the point at which the two graphs intersect.

Decreasing exponential graphs

9 Consider the two equations $y = 5^{-x}$ and $y = 6^{-x}$. Which is decreasing more rapidly for $x > 0$?

10 Consider the expression $y = 2^{-x}$.

i Complete the table of values:

x	-3	-2	-1	0	1	2	3
y							

ii Sketch the graph.

iii Describe what happens to the y-values as x increases.

iv Describe what happens to the y-values as x decreases.

v Does the curve cross the x-axis?

vi At what value of y does the graph cross the y-axis?

g Use your graph to approximate the solution of the equation $2^{-x} = 5$.

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11 For each of the following functions:

i Find the y-value of the y-intercept of the curve.

ii Complete the table of values:

x	-3	-2	-1	0	1	2	3
y							

iii Find the horizontal asymptote of the curve.

iv Sketch the graph.

a $y = 3^{-x}$ **b** $y = 4^{-x}$ **c** $y = 6^{-x}$

12 Consider the expression $y = 9^{-x}$.

i Can the value of y ever be negative?

ii Can the value of y ever be equal to 0?

iii As the values of x get larger and larger, what value does y approach?

iv As the values of x get smaller and smaller, what value does y approach?

v Find the y-value of the y-intercept of the curve.

vi How many x-intercepts does the curve have?

vii Sketch the graph.

h Use your graph to approximate the solution of the equation $9^{-x} = 5$.

13 Consider the exponentials $y = 5^{-x}$ and $y = 8^{-x}$:

a Find the horizontal asymptote of each curve:

i $y = 5^{-x}$ ii $y = 8^{-x}$

b Complete a table of values for each exponetial:

x	-3	-2	-1	0	1	2	3
y							

i $y = 5^{-x}$ ii $y = 8^{-x}$

c Sketch the graphs of $y = 5^{-x}$ and $y = 8^{-x}$ on the same set of axes.

d Find the coordinates of the point at which the two graphs intersect.

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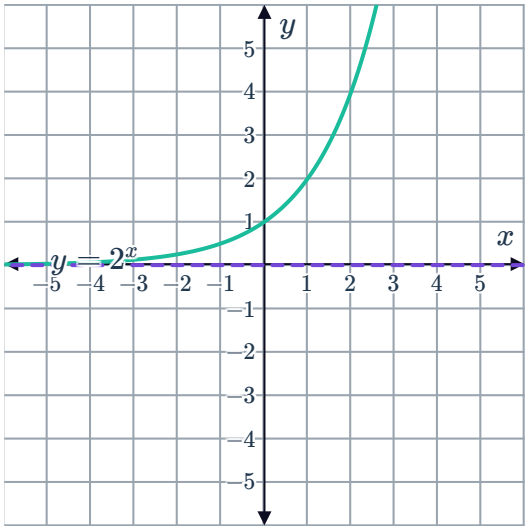
Dilations and reflections

14 How could we use the graph of $y = 7^x$ to draw the graph of $y = 7^{-x}$?

15 Consider the graph of $y = 2^x$:

a Describe a transformation of the graph of $y = 2^x$ that would obtain $y = 2^{-x}$.

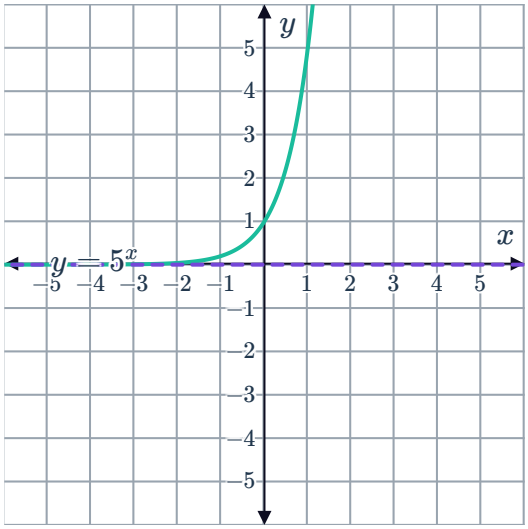
b Sketch the graph of $y = 2^{-x}$ on the same set of axes as $y = 2^x$.



16 Consider the graph of $y = 5^x$:

a Describe a transformation of the graph of $y = 5^x$ that would obtain $y = -5^x$.

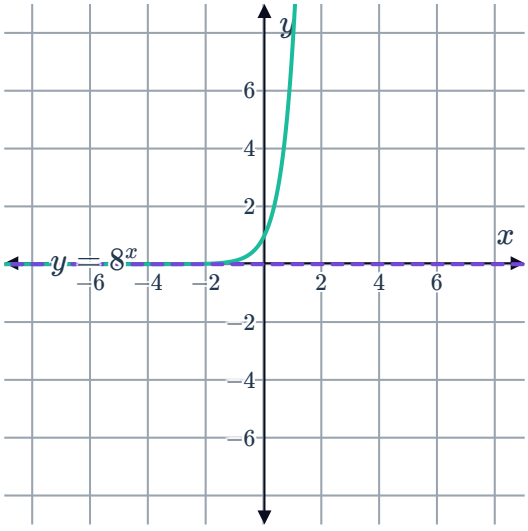
b Sketch the graph of $y = -5^x$ on the same set of axes as $y = 5^x$.



17 Consider the graph of $y = 8^x$:

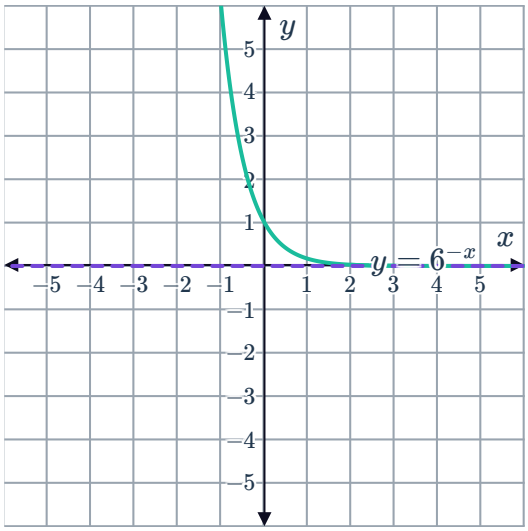
a Describe a transformation of the graph of $y = 8^x$ that would obtain $y = -8^{-x}$.

b Sketch the graph of $y = -8^{-x}$ on the same set of axes as $y = 8^x$



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18 Consider the graph of $y = 6^{-x}$:



a Describe a transformation of the graph of $y = 6^{-x}$ that would obtain $y = -6^{-x}$.

b Sketch the graph of $y = -6^{-x}$ on the same set of axes as $y = 6^{-x}$.

c Describe the graph of $y = -6^{-x}$ as x increases.

19 Of the two functions $y = 4^x$ and $y = \dfrac{4^x}{4}$, which is increasing more rapidly for $x > 0$?

20 Consider the function $y = 3\,(4^x)$.

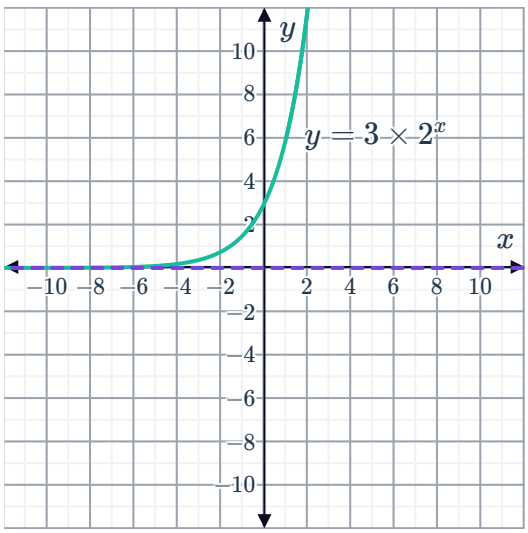
a Find the y-value of the y-intercept of the curve.

b Can the value of y ever be negative?

c As x approaches infinity, what value does y approach?

d Sketch the graph of $y = 3\,(4^x)$.

21 Consider the function $y = 3 \times 2^x$.



a Find the value of y when $x = 0$.

b Describe a transformation of the graph of $y = 3 \times 2^x$ that would obtain $y = -3 \times 2^x$.

c Sketch the graph of $y = -3 \times 2^x$ on the same set of axes as $y = 3 \times 2^x$.

d The number of bacteria over time is to be modelled by an exponential function, with x representing time and y representing the number of bacteria.

If the bacteria are increasing, which function should be used?

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22 Consider the original graph $y = 2^x$. The function values of the graph are multiplied by 3 to form a new graph.

a Complete the following table by finding the points on the new graph:

Point on original graph	$\left(-1, \frac{1}{2}\right)$	$(0, 1)$	$(1, 2)$	$(2, 4)$
Point on new graph	$(-1, \square)$	$(0, \square)$	$(1, \square)$	$(2, \square)$

b Find the equation of the new graph.

c Sketch the graphs of the original and new function on the same set of axes.

d For positive x-values, is the graph of the new function above or below the graph of 2^x?

e For negative x-values, is the graph of the new function above or below the graph of 2^x?

Translations

23 Consider the graph of $y = 6^x$:

a Find the y-value of the y-intercept of $y = 6^x$.

b Hence, find the y-value of the y-intercept of $y = 6^x + 3$.

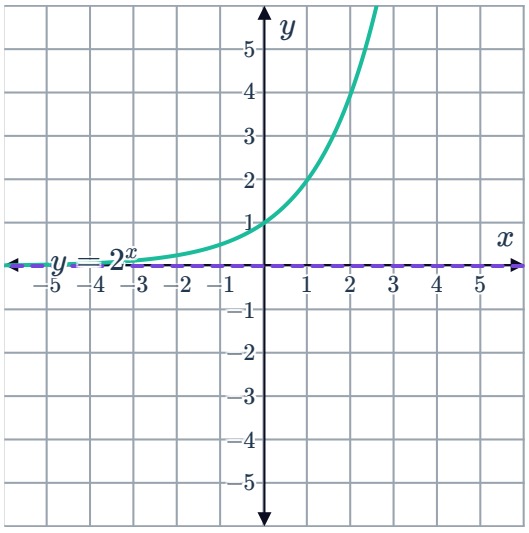
c Find the horizontal asymptote of $y = 6^x$.

d Hence, find the horizontal asymptote of $y = 6^x + 3$.

24 Consider the graph of $y = 2^x$:

a Describe the translation required to shift the graph of $y = 2^x$ to obtain the graph of $y = 2^x - 5$.

b Sketch the graph of $y = 2^x - 5$ on the same set of axes as $y = 2^x$.

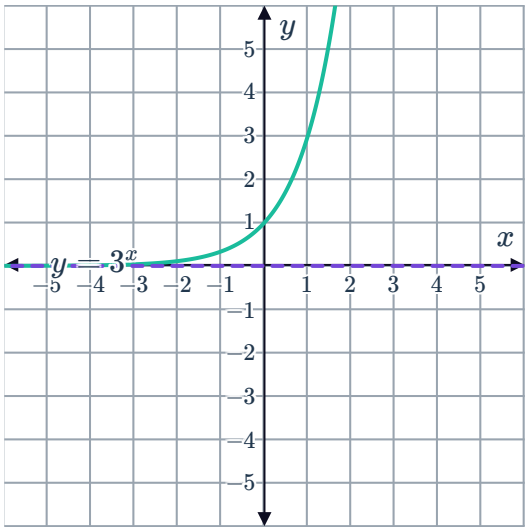


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25 Consider the graph of $y = 3^x$:

a Describe the translation required to shift the graph of $y = 3^x$ to obtain the graph of $y = 3^x + 1$.

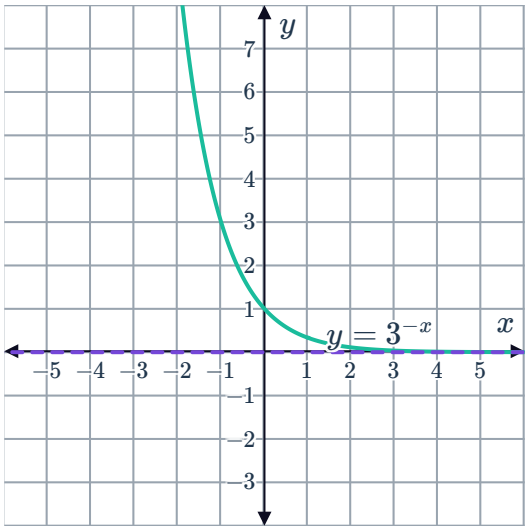
b Sketch the graph of $y = 3^x + 1$ on the same set of axes as $y = 3^x$.



26 Consider the graph of $y = 3^{-x}$:

a Describe the translation required to shift the graph of $y = 3^{-x}$ to obtain the graph of $y = 3^{-x} + 5$.

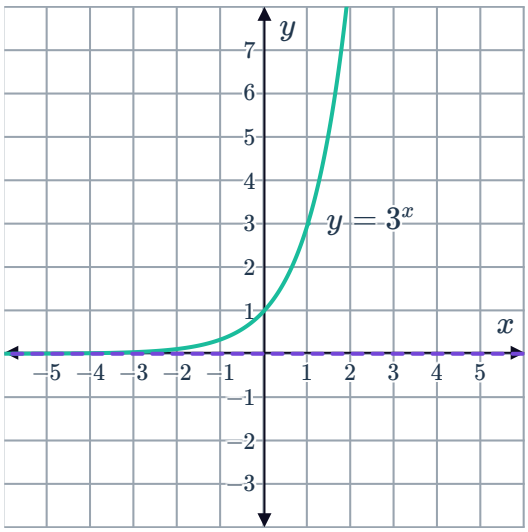
b Sketch the graph of $y = 3^{-x} + 5$ on the same set of axes as $y = 3^{-x}$.



27 Consider the graph of $y = 3^x$:

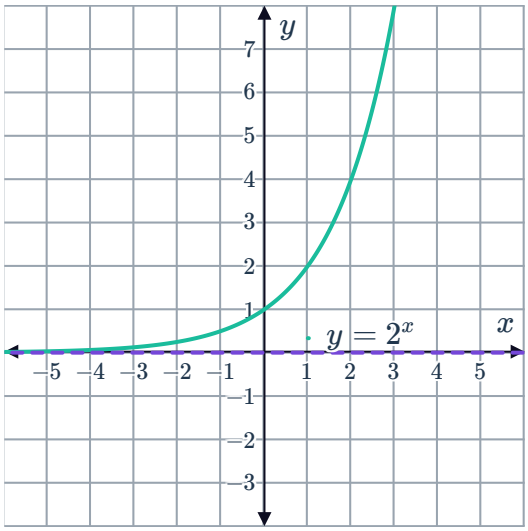
a Describe the translation required to shift the graph of $y = 3^x$ to obtain the graph of $y = 3^{x-2}$.

b Sketch the graph of $y = 3^{x-2}$ on the same set of axes as $y = 3^x$.



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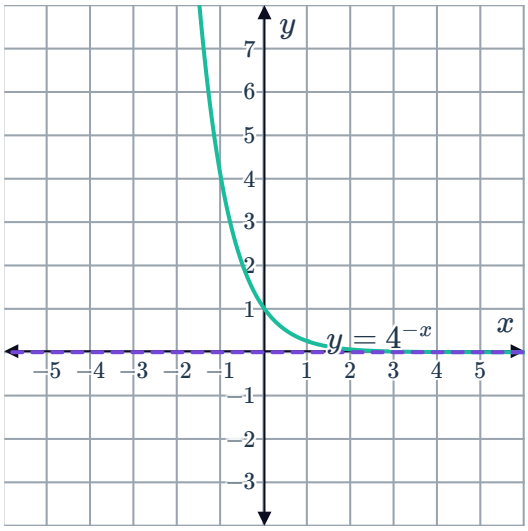
28 Consider the graph of $y = 2^x$:



a Describe the translation required to shift the graph of $y = 2^x$ to obtain the graph of $y = 2^{x+3}$.

b Sketch the graph of $y = 2^{x+3}$ on the same set of axes as $y = 2^x$.

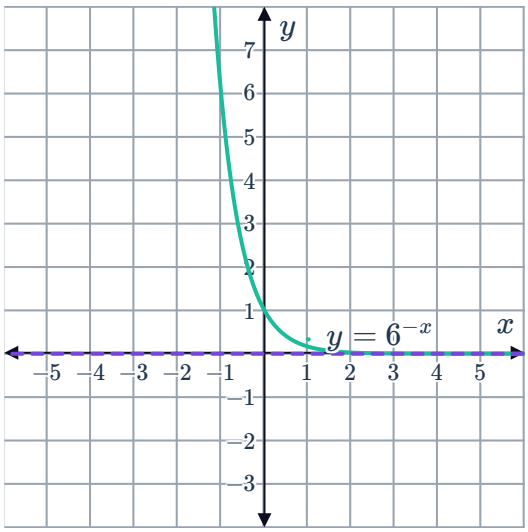
29 Consider the graph of $y = 4^{-x}$:



a Describe the translation required to shift the graph of $y = 4^{-x}$ to obtain the graph of $y = 4^{-x+1}$.

b Sketch the graph of $y = 4^{-x+1}$ on the same set of axes as $y = 4^{-x}$.

30 Consider the graph of $y = 6^{-x}$:



a Describe the translation required to shift the graph of $y = 6^{-x}$ to obtain the graph of $y = 6^{-x-4}$.

b Sketch the graph of $y = 6^{-x-4}$ on the same set of axes as $y = 6^{-x}$.

31 State the new equation when the graph of $y = 2^x$ is moved down by 9 units.

32 State the new equation when the graph of $y = 12^x$ is moved left by 6 units.

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33 State the new equation when the graph of $y = 3^{-x}$ is moved right by 2 units and down by 7 units.

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